

OVERVIEW

CervixScan is an open-source deep learning system that classifies cervical colposcopy images into Type 1, 2, or 3 transformation zones, helping clinicians and researchers make faster, more informed decisions. A finetuned-trained VGG16 CNN is served via a Flask REST API and accessed through a responsive web application.

KEY CAPABILITIES

- 3-class cervical type classification
- Real-time inference via REST API POST /classify
- Drag-and-drop image upload with live preview
- Per-class softmax confidence probability
- Lightweight TFLite model

DATASET

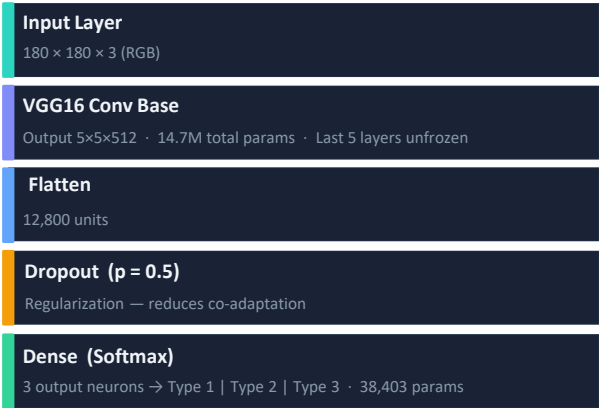
Intel MobileODT Cervical Cancer Dataset

DATA PROCESSING

- Resize all images to 180 × 180 pixels (RGB)
 - Normalize pixel values to range [0.0, 1.0]
 - Detect & remove corrupt / truncated image files
 - Stratified train / val / test split (80% / 10% / 10%)
- Rotation ±40° · Zoom 20% · Width shift 20%
· Shear 20% · Horizontal & Vertical flip enabled

MODEL ARCHITECTURE

VGG16 Base (ImageNet weights); Custom Softmax Head



Training Configurations

Optimizer	Adam (learning rate = 0.0001)
Loss	Sparse Categorical Cross-Entropy
Batch Size	32
Max Epochs	100 (early stopped at epoch 52)
LR Schedule	ReduceLROnPlateau (factor 0.2, patience 10)
Early Stop	Patience 20, restore best weights
Checkpoint	ModelCheckpoint (monitors val_loss)

TRAINING & EVALUATION RESULTS



DEPLOYMENT

GitHub Repository
Full source code including model training notebook, Flask API backend, and responsive frontend. Open for inspection and reproducibility.
github.com/Samuel-Ozechi/CervixScan

Web Application
Responsive SPA with drag-and-drop upload, live image preview, real-time classification, and per-class animated confidence bars.
cervixscan.com

IMPACT

Critical Screening Gap in Sub-Saharan Africa
Cervical cancer is the leading cause of cancer-related death among women in Sub-Saharan Africa. Providing automated diagnostics at the point of care specialists and makes essential screening accessible.

Academic Citation · Peer-Reviewed Publication
This project was cited in the AIP Conference Proceedings , Vol. 3220, Article 030002 · American Institute of Physics
"A regularized CNN approach for detecting cervical cancer"
<https://pubs.aip.org/aip/acp/article-abstract/3220/1/030002/3315899/>